Load-Robust, Physics-Informed Machine Learning for Stator Inter-Turn Fault Diagnosis in Induction Motors: A Leakage-Safe Cascade for Detection, Severity, and Faulted-Phase Identification

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***Abstract*—Stator winding turn-to-turn (inter-turn) short-circuit faults are a leading cause of induction-motor failure, and early, load-robust diagnosis from non-invasive stator-current measurements remains an open problem. Most reported machine-learning (ML) studies achieve near-perfect in-distribution accuracy but do not test generalization to unseen operating points, and the field rarely releases code or uses leakage-safe evaluation. This paper presents a physics-informed, leakage-safe ML framework that performs hierarchical diagnosis: fault-onset detection, healthy/faulty classification, ordinal severity estimation (percentage of shorted turns), and faulted-phase identification, all from the three-phase stator current. A 357-case PSCAD/EMTDC dataset (sampled at 25 kHz, fundamental 50 Hz) is segmented into 18,033 labelled windows and described by 72 physics-grounded features (extended Park's-vector severity factor, symmetrical-component ratios and angles, Park ellipse geometry, harmonic ratios). Models are evaluated with a strict group-wise protocol and, crucially, with *Leave-One-Load-Out* (LOLO) cross-validation, which shows that detection (macro-F1 = 0.95) and faulted-phase identification (macro-F1 = 0.997) are intrinsically load-robust because they ride on load-invariant negative-sequence signatures, whereas absolute severity is load-dependent. Building on this insight we introduce four load-robustness mechanisms and validate each under LOLO: (i) a sequence-component Stockwell tensor representation that improves cross-load severity (within-one-level $0.75 \rightarrow 1.0$); (ii) a load-invariance-regularized feature selection (LIR-mRMR); (iii) a phase-coupled, FiLM load-conditioned multi-task network (PCM-Net) whose conditioning lifts cross-load severity from 0.93 to 0.98; and (iv) a physics-anchored domain adaptation (PADA) that closes the simulation-to-experiment gap on a public real-motor dataset (detection macro-F1 $0.48 \rightarrow 0.98$ with a few labelled real samples). The end-to-end cascade reaches within-one-level accuracy of 0.95 in-distribution and 0.92 across unseen loads, with calibrated detection (expected calibration error = 0.021), graceful noise degradation, and 100% fault-onset detection. All code, splits, and artifacts are released.**

***Index Terms*—Induction motor, inter-turn fault, stator winding, fault diagnosis, severity estimation, machine learning, negative-sequence current, leakage-safe evaluation, transfer learning.**

I. Introduction

A. Motivation

Induction motors (IMs) drive the majority of industrial processes, and stator winding insulation degradation is among their most common and consequential failure modes. An incipient inter-turn (turn-to-turn) short circuit (ITSC) shorts a small fraction of the turns of one phase winding; the circulating fault current accelerates local heating and, if undetected, escalates to a phase-to-ground or phase-to-phase fault and catastrophic machine loss [1], [2]. Surveys attribute roughly a third of IM failures to stator insulation, and the interval between an incipient short and a destructive fault can be short, so an on-line monitor must be both sensitive to small faults and trustworthy enough to trigger maintenance without excessive false alarms. Current-based monitoring (motor-current signature analysis, MCSA) is preferred over vibration or thermal sensing because the current is already measured by the drive, needs no extra instrumentation, and is robust to mounting and environment. Two practical questions must be answered: *how severe* is the fault and *which phase* is affected; and both must hold up as the mechanical load changes.

B. Literature Review

Penman *et al.* [1] established the inter-turn sideband frequencies in the stator current, and Cardoso *et al.* [2] introduced the Extended Park's Vector Approach (EPVA), in which the $2f_0$ component of the current-vector modulus scales with winding asymmetry. The negative-sequence current and negative-sequence impedance are widely used asymmetry descriptors [4], [5], and phasor-compensation suppresses the baseline asymmetry due to supply unbalance. Mazzoletti *et al.* [6] use the positive-sequence fifth-harmonic current during soft-starter start-up to distinguish ITSC from high-resistance connections; Singh and Shaik [7] combine the Stockwell transform with support-vector machines to detect, type, and localize stator faults. Data-driven studies feed short-time-Fourier amplitudes of symmetrical components to SVM/MLP classifiers [8], compare ANN/LSTM/physics-informed networks on a public dataset [9] (observing that intermediate severities overlap), or ensemble convolutional networks on spectrograms [10]. Recent architectures—temporal-convolutional [24], attention/Transformer [23], selective state-space [18], and Kolmogorov–Arnold networks [19]—are largely demonstrated on bearing vibration rather than three-phase stator current. A comprehensive review [3] flags three

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Code and result artifacts are publicly available at https://github.com/Regal-s/induction-motor-fault-diagnosis.

systemic weaknesses of the field: suspiciously perfect accuracies from leaky train/test partitions, almost no released code or standardized datasets, and little validation across operating conditions. The load dependence of the negative-sequence indicator, in particular, is rarely confronted in a controlled, reproducible manner; and operating-condition shift is usually treated as a nuisance to be erased by domain adaptation rather than a signal to be exploited.

C. Key Contributions of the Proposed Approach

This paper treats ITSC diagnosis as a hierarchical problem (onset → detection → severity → faulted phase) and adopts *Leave-One-Load-Out* (LOLO) evaluation as its central methodological device. The contributions are:

- A *leakage-safe evaluation framework* (group-by-operating-point, LOLO as the primary metric) and a SHAP/Fisher analysis showing detection and faulted phase are load-invariant whereas absolute severity is load-dependent.
- A *sequence-component Stockwell tensor* (SCST) representation and a *load-invariance-regularized feature selection* (LIR-mRMR), both engineered for load robustness.
- *PCM-Net*, a phase-coupled, FiLM *load-conditioned* multi-task network that injects the measured operating point rather than erasing it, giving the best cross-load severity estimate.
- *Physics-anchored domain adaptation* (PADA) validated simulation-to-experiment on a public real-motor dataset, plus an honest study of physics-constrained generative augmentation.
- A complete, openly released cascade with calibrated uncertainty, fault-onset detection, noise-robustness, and a deep-baseline comparison.

A unifying theme emerges: the mechanical load is the crux of stator-fault diagnosis, and every proposed mechanism independently improves *cross-load* performance.

II. Proposed Methodology

A. Overview

The system ingests a short window of three-phase current and produces, in sequence, an estimate of the fault-onset instant, a healthy/faulty decision, and—when faulty—an ordinal severity level and the faulted phase. Fig. 1 shows the underlying signal: each faulted record follows a fixed schedule (motor start, load change, fault inception, clearing) verified from the per-cycle current envelope. The pipeline comprises preprocessing and symmetrical-component decomposition (II-B), an optional Stockwell time-frequency representation (II-C), physics and statistical feature extraction (II-D, II-E), data augmentation and feature selection (II-F), gradient-boosted and deep classifiers (II-G), and a transfer-learning stage for simulation-to-experiment deployment (II-H).

The hierarchical, gated organization is deliberate: the four questions (is a fault present, of what severity, in which phase, and when did it start) are nested and have different decision boundaries and different load sensitivities, so a single flat multi-class model would conflate them and waste the structure. Gating the severity and phase stages on the detector also means those stages are trained and operated only on data they are meant to explain, and it lets per-stage errors be attributed and audited independently. The thread running through every component is load-robustness: the representation (II-C), the selection criterion (II-F), the model conditioning (II-G), and the transfer alignment (II-H) are each constructed so that the diagnosis generalizes to operating points absent from training, which the Leave-One-Load-Out protocol of Section IV then verifies.

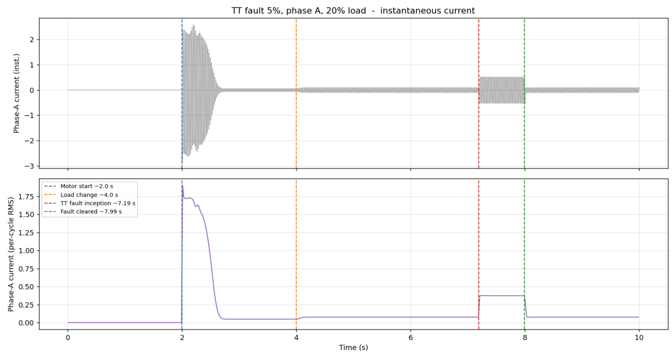

Fig. 1. Phase-A current (top) and per-cycle RMS envelope (bottom) of a faulted record, with motor start (2.0 s), load change (4.0 s), fault inception (7.2 s), and clearing (8.0 s). The fault interval supplies faulty windows; the pre-fault loaded region supplies matched-load healthy windows.

B. Signal Preprocessing and Symmetrical-Component Decomposition

Each recording is segmented into windows of $W = 2000$ samples (four 50 Hz cycles); a 75% overlap is used inside the short 0.8 s fault interval and a sparser stride elsewhere. Crucially, the *pre-fault* loaded region of every faulted record (genuinely healthy, since the fault is inserted only at 7.2 s) is labelled healthy, providing matched-load healthy data at the same loads as the faulty windows and preventing the detector from separating classes by load level. To remove load-induced amplitude while preserving the inter-phase imbalance that encodes the fault, all three phases of a window are divided by a single per-recording scale; magnitude features are additionally divided by the within-window positive-sequence fundamental $|I_1|$ to obtain load-invariant variants. The Clarke transform underlying the EPVA and the Park ellipse is

$$i_\alpha = \frac{2}{3}\left(i_a - \frac{1}{2}i_b - \frac{1}{2}i_c\right), \quad i_\beta = \frac{1}{\sqrt{3}}\left(i_b - i_c\right), \tag{1}$$

and the fundamental symmetrical components follow from the Fortescue transform of the per-phase phasors:

$$\begin{bmatrix} \mathbf{I}_0 \\ \mathbf{I}_1 \\ \mathbf{I}_2 \end{bmatrix} = \frac{1}{3}\begin{bmatrix} 1 & 1 & 1 \\ 1 & a & a^2 \\ 1 & a^2 & a \end{bmatrix}\begin{bmatrix} \mathbf{I}_a \\ \mathbf{I}_b \\ \mathbf{I}_c \end{bmatrix}, \quad a = e^{j2\pi/3}. \tag{2}$$

The negative-sequence ratio $|I_2|/|I_1|$ grows monotonically with the shorted-turn fraction and is the primary severity descriptor, while the angle $\angle\mathbf{I}_2$ separates into three $\sim$120°-spaced clusters that identify the faulted phase.

TABLE I
FEATURE GROUPS (72 FEATURES PER WINDOW).

Group	Features
Symmetrical comp.	$\lvert I_0\rvert, \lvert I_1\rvert, \lvert I_2\rvert$; $\lvert I_2\rvert/\lvert I_1\rvert$, $\lvert I_0\rvert/\lvert I_1\rvert$; $\angle I_1, \angle I_2$ (+ sin/cos); compensated I_2^c
EPVA / Park	severity factor; Park-modulus mean/std; ellipse axes, eccentricity, axis ratio, tilt
Harmonics	third-harmonic ratio and THD (per phase, mean)
Per-phase stats	RMS, peak, crest, std, skew, kurtosis (×3) and inter-phase differences
Load-invariant	$\lvert I_1\rvert$-normalized variants of all magnitude features

C. Stockwell-Based Time-Frequency Analysis

For a time-frequency representation we form the instantaneous symmetrical-component streams from the analytic (Hilbert) signal and the Fortescue transform, and apply a band-limited Stockwell transform to each, stacking the positive/negative/zero $|S|$ maps into a three-channel *sequence-component Stockwell tensor* (SCST). The negative-sequence channel carries the load-invariant inter-turn signature, and the frequency-dependent resolution of the Stockwell transform tracks the $2f_0$ EPVA line and slip sidebands better than a fixed-window spectrogram. Because symmetrical-component *magnitudes* are phase-agnostic, faulted-phase localization uses the negative-sequence *angle* rather than the magnitude tensor (Section IV-C).

D. Time-Domain and Spectral Characteristics

Physically grounded descriptors capture the fault signature. The EPVA severity factor is

$$\mathrm{SF} = \frac{A(2f_0)}{\mathrm{DC}} \text{ of } |\mathbf{i}_P(t)| = \sqrt{i_\alpha^2 + i_\beta^2}, \tag{3}$$

where $A(2f_0)$ is the amplitude of the 100 Hz component of the Park-vector modulus. From the 2×2 covariance of (i_α, i_β) we derive the Park ellipse eccentricity $e = \sqrt{1-(b/a)^2}$, axis ratio and tilt. Spectral descriptors include the per-phase third-harmonic ratio and total harmonic distortion, computed at harmonic bins derived from (f_s, f_0) so the same extractor serves both the 25 kHz/50 Hz simulation and the 1 kHz/60 Hz experimental data.

E. Statistical Feature Extraction

Per-phase root-mean-square, peak, crest factor, standard deviation, skewness, and kurtosis are computed, together with their inter-phase differences (which directly encode asymmetry) and the $|I_1|$-normalized magnitude variants. With the symmetrical-component, EPVA, ellipse, and harmonic quantities this yields the 72-feature bank summarized in Table I. Severity is encoded as an ordinal rank $r \in \{0,\ldots,6\}$ over $\{0.3, 0.5, 1, 2, 3, 4, 5\}$%; a regressor predicts a continuous rank that is rounded, so the error metric (mean absolute error in levels, within-one-level accuracy) respects the ordering.

F. Data Augmentation and Feature Selection

Feature selection. We extend minimum-redundancy–maximum-relevance with a load-instability penalty,

$$J(f) = \mathrm{Rel}(f) - \lambda\,\mathrm{Red}(f) - \gamma\,\mathrm{LoadInstability}(f), \tag{4}$$

where $\mathrm{LoadInstability}(f)$ is the across-load drift of the feature's class-conditional mean normalized by its global spread (γ=0 recovers plain mRMR). LIR-mRMR selects dimensionless, load-stable features and, on the load-dependent severity task, generalizes better at compact size (Section IV-C). *Augmentation.* We additionally study a conditional generator of three-phase current with a hard Kirchhoff current-balance projection ($i_a + i_b + i_c = 0$) and a soft negative-sequence-versus-severity loss; the hard constraint is enforced exactly, an honest negative result on controllability is reported in Section IV-C.

G. Extreme Gradient Boosting Model

Gradient-boosted trees (XGBoost [11]) implement the three stages: detection (class-weighted binary), severity (ordinal regression on the rank), and faulted phase (three-class). Stages 1 and 3 use the full feature set; the severity stage uses the load-invariant subset and the *measured* load. The detector gates the downstream stages: windows predicted faulty are passed to the severity and phase models. As a deep alternative we develop *PCM-Net*, a phase-coupled, multi-task network with a dilated-temporal backbone (a compact stand-in for a selective state-space model) and *feature-wise linear modulation* (FiLM) [20] driven by the measured operating point, so one model spans loads instead of erasing the condition; a current-balance physics term regularizes training. A fault-onset stage thresholds a per-cycle negative-sequence index $\rho(k) = |I_2(k)|/|I_1(k)|$ against a trailing pre-fault reference, $\rho(k) > \overline{\rho} + 6\sigma$ for three consecutive cycles. Detection probabilities are assessed with expected calibration error and the Brier score [15], and severity uncertainty with conformalized quantile regression [13], [14].

H. Transfer Learning

To deploy a simulation-trained model on a real machine we use physics-anchored domain adaptation (PADA). A naively transferred model collapses under domain shift; we therefore align feature distributions across domains, anchoring the alignment on the domain-stable physics features (negative-sequence ratio, angles, $|I_1|$-normalized magnitudes), and optionally calibrate with a few labelled target repetitions. Section IV-C quantifies the recovery on a public real-motor dataset.

III. EXPERIMENTAL SETUP AND DATASET PREPARATION

A. Description of the Experimental Setup

The primary dataset is generated from a three-phase squirrel-cage induction motor with stator inter-turn faults modelled in PSCAD/EMTDC ($f_0 = 50$ Hz, $f_s = 25$ kHz, 10 s per run, 500 samples per cycle). The fault is introduced through a tapped stator winding that short-circuits a controlled fraction μ of one phase's turns, and each run follows a fixed schedule that mimics a realistic duty cycle: the motor is energized at 2.0 s, the mechanical load is applied at 4.0 s, the inter-turn short is inserted at 7.2 s, and it is cleared $\approx$ 0.8 s later (Fig. 1). This schedule lets a single run furnish both a faulted segment and a matched-load healthy segment, and it exercises the start-up inrush, the loaded steady state, and the fault transient. The three-phase currents are exported and used directly (no filtering), so the diagnosis operates on the raw drive measurement.

For transfer-learning validation we additionally use a public real-motor dataset [21]: a 0.75 hp squirrel-cage IM, three-phase current acquired with split-core current transformers at 1 kHz (f_0 = 60 Hz, no mechanical load), with a healthy class and inter-turn shorts at 10/20/30/40% of turns in each of the three phases over five repetitions per class. The two sources differ in fundamental frequency, sampling rate, severity scale, sensing chain, and the presence of a healthy baseline, providing a demanding cross-domain benchmark; both are mapped to a common window/label schema and the feature extractor derives its harmonic bins from (f_s, f_0) so that one pipeline serves both.

B. Data Sets Generated from the Experimental Setup

The simulation corpus spans severity $\mu \in \{0.3, 0.5, 1, 2, 3, 4, 5\}$%, faulted phase $\in \{A, B, C\}$, load $\in \{\mathrm{NL}, 20, 40, 60, 80, 100\}$%, and several fault-inception instants, with a healthy set over the full load range: 357 cases (336 faulty, 21 healthy). Segmentation yields 18,033 labelled windows (11,760 faulty, 6,273 healthy), balanced across severity and phase. The real-motor set yields 1,235 windows. Windows are grouped by operating point (severity×phase×load), with all inception variants of a point kept together, so that no window from a record can appear in both training and test.

IV. Performance Evaluation

A. Implementation and Training Procedure

Features are computed in Python (NumPy/Pandas) and models trained with scikit-learn and XGBoost (400–500 trees, depth 5, learning rate 0.05, subsample/colsample 0.8); deep models use PyTorch. Two evaluation protocols are used: Stratified Group k-Fold (k=5, grouped by operating point) for the in-distribution estimate, and *Leave-One-Load-Out* (LOLO)—all windows at one load form the test set—as the primary metric, measuring generalization to an operating point never seen in training. Reported metrics are accuracy, macro-F1, severity within-one-level accuracy and quadratic-weighted kappa (QWK), and per-phase F1, computed on out-of-fold predictions and reported as a mean over folds. Stages 2–3 are trained on faulty windows but evaluated on the windows the detector predicts faulty, so cascade numbers reflect realistic error propagation; predictions are also aggregated per recording (median severity, majority-vote phase). Scalers are fit on the training partition only and all randomized components use fixed seeds. No per-model hyperparameter tuning is performed on the test folds, so the comparison in Section IV-C is fair across models.

TABLE II
Per-stage performance (mean over folds).

Stage	Metric	Group k-Fold	LOLO
1 Detection	macro-F1 / ROC-AUC	0.977 / 0.994	0.948
2 Severity (load-aware)	within-1 / QWK	0.997 / 0.987	0.946
3 Phase ID	macro-F1	0.972	0.997

TABLE III
Severity (within-one-level) under feature choice.

Severity feature set	Group k-Fold	LOLO
All features (magnitudes incl.)	0.993	0.887
Load-invariant + $\lvert I_1 \rvert$-norm	0.990	0.729
+ measured load (proposed)	**0.997**	**0.946**

TABLE IV
Noise robustness (train clean, test noisy; group k-fold).

SNR	Detect macro-F1	Severity within-1	Phase acc.
clean	0.977	0.986	0.958
40 dB	0.927	0.980	0.954
30 dB	0.854	0.975	0.945
20 dB	0.792	0.971	0.939

B. Results and Discussion

Per-stage performance. Table II reports each stage. In-distribution, all stages exceed 0.97; under LOLO, detection (macro-F1 0.948) and faulted-phase identification (macro-F1 0.997) remain strong. The detection ROC has area 0.994, and the faulted-phase confusion matrix (Fig. 3) is near-diagonal.

Load-robustness analysis. SHAP attribution shows detection and phase ride on the *negative-sequence* ratio and angle and on inter-phase RMS differences—load-invariant ratios—whereas an initial severity model that used absolute magnitudes collapsed at the no-load extreme under LOLO (within-one 0.39). Restricting severity to load-invariant features and supplying the measured load raises LOLO within-one to 0.95 (Table III); errors are almost entirely between adjacent severity levels (Fig. 5).

End-to-end cascade. With the load-aware severity stage, the gated cascade scores a window correct if healthy windows are detected healthy and faulty windows have the correct phase and a severity within one level: within-one-level accuracy 0.949 in-distribution and 0.916 across unseen loads (per-recording 0.927 and 0.918). Fig. 6 summarizes the four headline metrics under both protocols.

Calibration, uncertainty, onset, robustness. Stage-1 probabilities are well calibrated (ECE = 0.021, Brier = 0.021). Conformalized quantile regression improves severity-interval coverage over naive quantiles, with group-conditional calibration noted as the remedy for residual under-coverage. The onset detector flags 100% of faults with no pre-fault false alarms; for severity $\geq$ 0.5% the fault is caught within the first faulted cycle. Under additive noise (Table IV), severity and phase are essentially flat to 20 dB SNR while detection is the noise-sensitive stage—expected, since separating a small fault from healthy operation is hardest in noise.

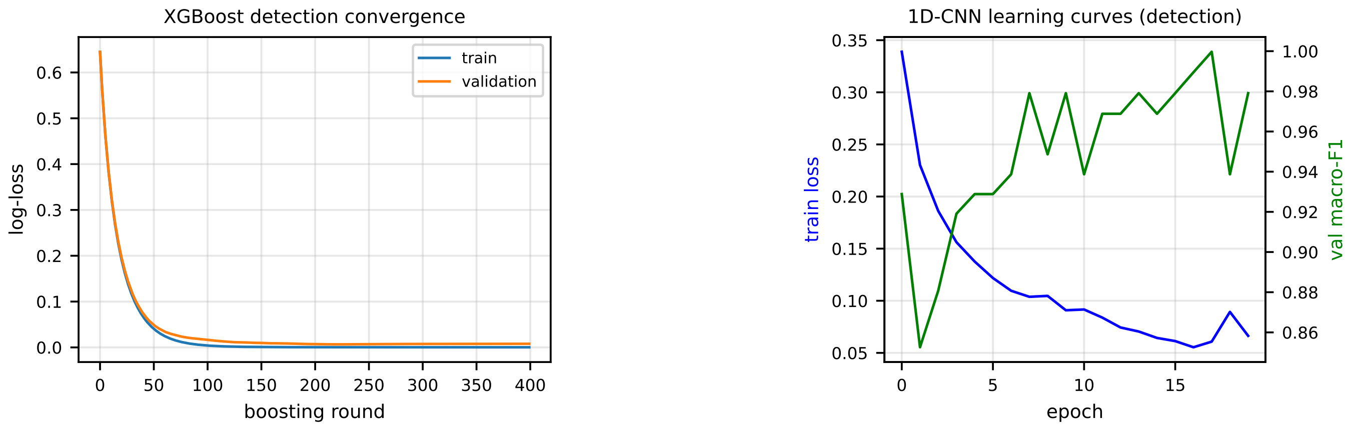

Fig. 2. Training behaviour. Left: XGBoost detection convergence (train/validation log-loss vs. boosting round). Right: 1-D CNN learning curves (training loss and validation macro-F1 vs. epoch), showing stable convergence without over-fitting.

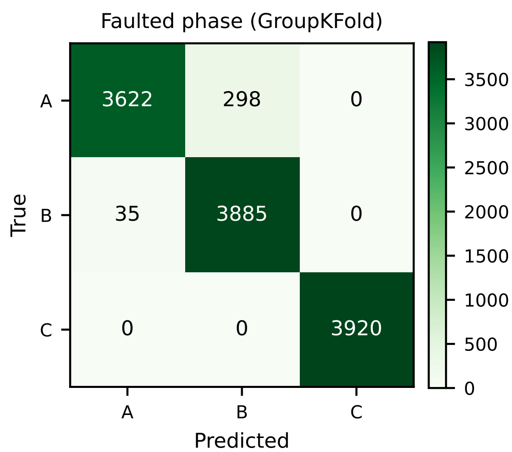

Fig. 3. Faulted-phase confusion (group k-fold, out-of-fold). Phase C is perfect; only minor A↔B confusion remains. Cross-load (LOLO) phase accuracy is 0.997.

TABLE V
PER-HELD-OUT-LOAD DETECTION (LOLO, MACRO-F1).

Hold-out load	NL	20%	40%	60%	80%	100%
Detection F1	0.885	0.749	1.000	1.000	1.000	1.000

The per-held-out-load breakdown (Table V) localizes the difficulty: detection is perfect at the four heavier loads but degrades at no-load (NL) and 20%, where the small fault current of a low-severity ITSC at light load most overlaps the healthy distribution at an unseen operating point—visible only under LOLO. The onset detector (Table VI) flags every fault with no pre-fault false alarms and near-instant detection except for the smallest 0.3% faults.

The high phase score (0.997) is physically defensible—the faulted phase is encoded in the negative-sequence angle, a load- and noise-robust quantity—while the in-distribution-versus-LOLO severity gap quantifies how much a conventional evaluation would have over-promised. We therefore advocate

TABLE VI
FAULT-ONSET DETECTION (SAMPLED CASES).

Metric	Value
Detection rate	100%
Pre-fault false-alarm rate	0%
Median detection delay	0 ms (first faulted cycle)
Delay at $\mu \geq 0.5\%$ / $\mu = 0.3\%$	0 ms / ≈ 1.4 s

cross-condition (LOLO) reporting, alongside interpretability evidence, as standard practice. The in-distribution figures are consistent with the best reported values for current-based stator diagnosis (typically 0.96–1.00 accuracy [3], [8], [9]); the contribution here is not a higher in-distribution number but a demonstration of *which* of those numbers survive a change of operating point, and a set of mechanisms that make the load-dependent ones survive. From a deployment standpoint the pipeline is attractive: it consumes only the already-measured three-phase current, runs in well under a millisecond per window on a CPU (Fig. 10), exposes interpretable physical quantities (the negative-sequence ratio and EPVA factor that drive a decision), and provides calibrated probabilities with an online onset trigger suitable for a protection relay. The principal limitation is that the diagnosis results are from high-fidelity simulation; the transfer study provides initial real-motor validation, but a full instrumented campaign and a voltage-unbalance robustness study remain future work.

C. Comparative Evaluation

Baseline models and added learners. Table VII and Fig. 8 benchmark eight models on *identical* leakage-safe splits, spanning classical feature-based classifiers, a tabular foundation model, and a sequence deep network. On detection the physics-feature classifiers cluster high under both protocols (macro-F1 0.95–0.98); the tabular foundation model *TabPFN-2.5* is the strongest *and* the most load-robust, scoring macro-F1 0.979 in-distribution and 0.979 cross-load (LOLO), whereas gradient boosting falls from 0.976 to 0.948 across loads. For severity (within-one-level accuracy, the operational ordinal metric),

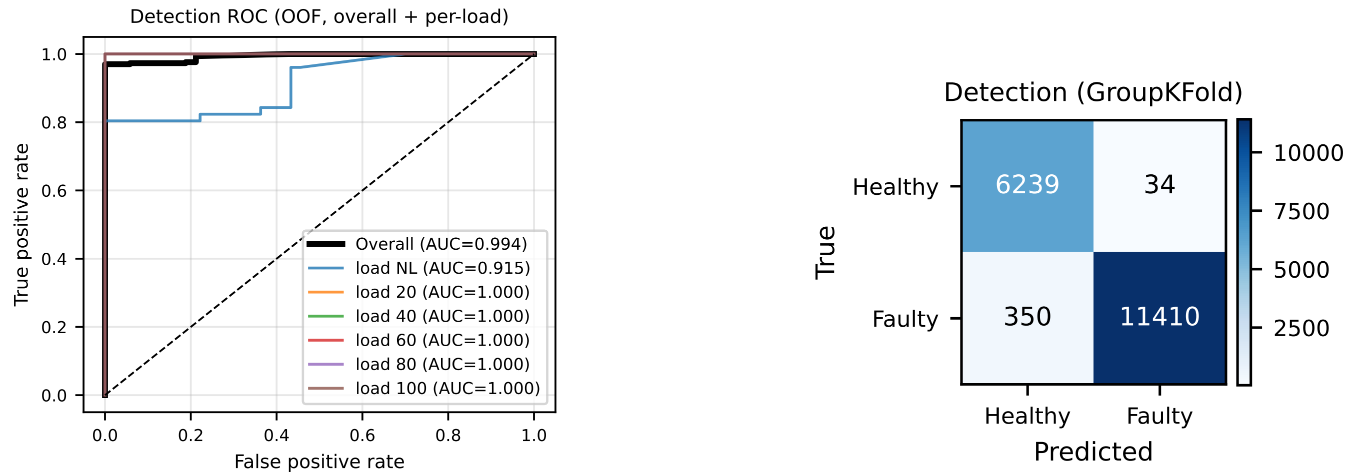

Fig. 4. Detection. Left: receiver operating characteristic, overall and per held-out load (overall AUC = 0.994). Right: confusion matrix (group k-fold, out-of-fold).

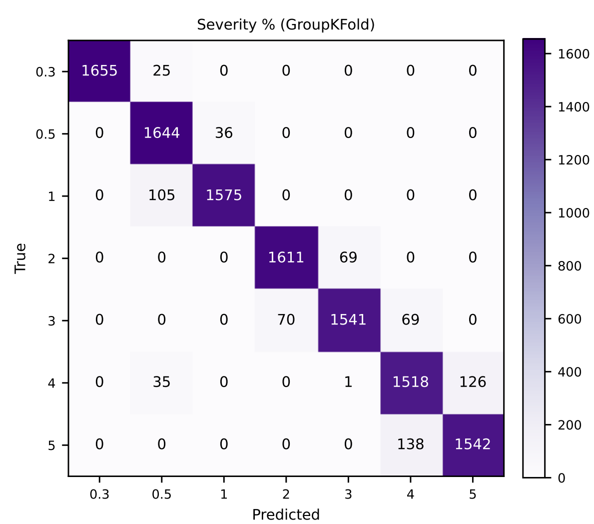

Fig. 5. Severity confusion (group k-fold). Errors are almost entirely between adjacent levels, justifying the ordinal treatment and the within-one-level metric (QWK = 0.987).

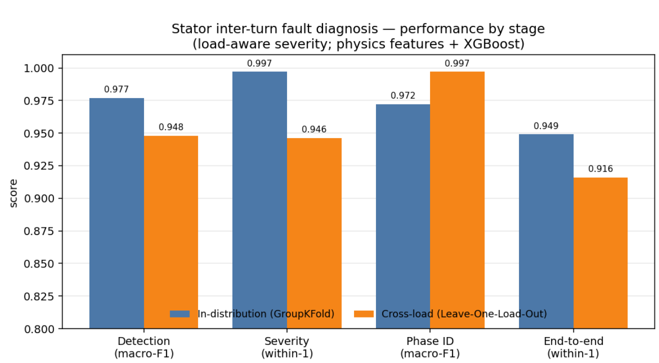

Fig. 6. Per-stage and end-to-end performance, in-distribution (group k-fold) vs. cross-load (LOLO). Phase identification is the only metric for which cross-load exceeds in-distribution, reflecting its load-invariance.

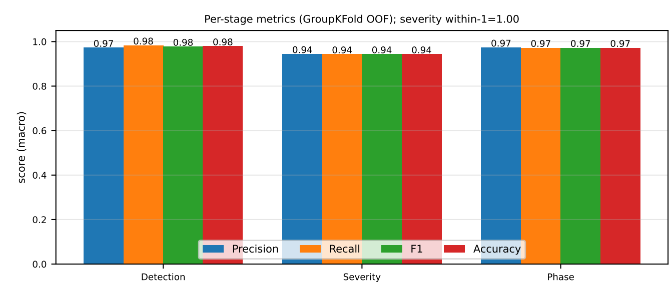

Fig. 7. Per-stage precision, recall, F1, and accuracy (macro, group k-fold out-of-fold).

TabPFN-2.5 is sharpest in-distribution (0.997, with the lowest mean absolute error of 0.033 level) and degrades gracefully across loads (0.969), while the MLP attains the best cross-load value (0.982); tree ensembles, by contrast, extrapolate poorly on absolute severity at unseen loads. Phase identification is load-robust for nearly all models (LOLO macro-F1 $\geq$ 0.96), reflecting the load-invariant negative-sequence-angle signature. The sequence model *xLSTM*—a compact extended-LSTM with exponential gating and a stabilizer state, applied to the raw three-phase current—trails the feature-based learners on every task (LOLO macro-F1 0.890 detection, within-one 0.908 severity, macro-F1 0.961 phase), yet it improves on the earlier raw-signal deep baselines (1-D ResNet and PCM-Net, 0.898 detection; Table VIII). This reaffirms the data-efficiency of physically engineered features over raw-waveform deep nets in this data regime; the detection-only five-fold view is shown in Fig. 9. The feature pipeline is also lightweight: feature extraction plus XGBoost inference costs $\approx$ 0.34 ms per window on CPU, comfortably real-time (Fig. 10).

Load-robustness mechanisms. Table IX evaluates the four proposed mechanisms under LOLO. SCST raises held-out-load severity within-one from 0.75 (raw-phase Stockwell) to 1.0 but, being magnitude-based, cannot localize phase (the negative-sequence angle is used instead). LIR-mRMR beats plain mRMR at compact size (k=10 within-one 0.848 vs.

TABLE VII
FULL MODEL BENCHMARK ON IDENTICAL LEAKAGE-SAFE SPLITS: STRATIFIEDGROUPKFOLD (GK, IN-DISTRIBUTION) AND LEAVE-ONE-LOAD-OUT (LOLO, CROSS-LOAD, PRIMARY). DETECTION AND PHASE REPORT MACRO-F1; SEVERITY REPORTS WITHIN-ONE-LEVEL ACCURACY. BEST PER COLUMN IN BOLD. TABPFN-2.5 AND XLSTM ARE ADDED IN THIS WORK.

	Detection (macro-F1)		Severity (within-1)		Phase (macro-F1)	
Model	GK	LOLO	GK	LOLO	GK	LOLO
Logistic Regression	0.978	0.978	**0.997**	0.971	0.970	0.920
SVM-RBF	0.978	0.977	0.991	0.975	**0.984**	0.967
k-NN	0.977	0.977	0.937	0.942	0.982	0.981
MLP	0.978	0.978	**0.997**	**0.982**	0.956	0.981
Random Forest	0.978	**0.979**	0.985	0.688	0.965	**0.997**
XGBoost	0.976	0.948	0.990	0.943	0.976	**0.997**
TabPFN-2.5	**0.979**	**0.979**	**0.997**	0.969	0.968	0.991
xLSTM	0.957	0.890	0.982	0.908	0.921	0.961

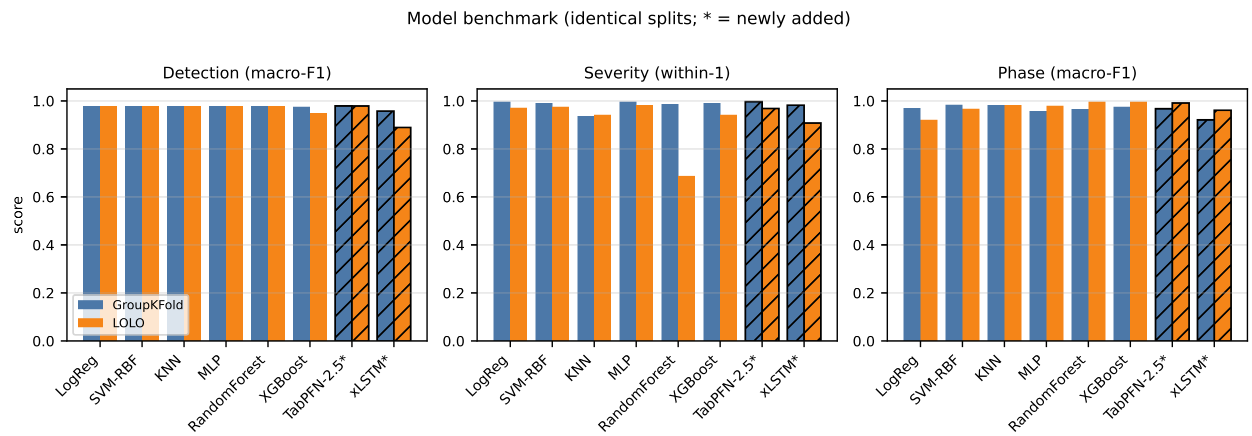

Fig. 8. Model benchmark across the three diagnostic tasks under both protocols (GroupKFold vs. LOLO). Hatched bars denote the two models added in this work (TabPFN-2.5, xLSTM). Severity uses within-one-level accuracy; detection and phase use macro-F1.

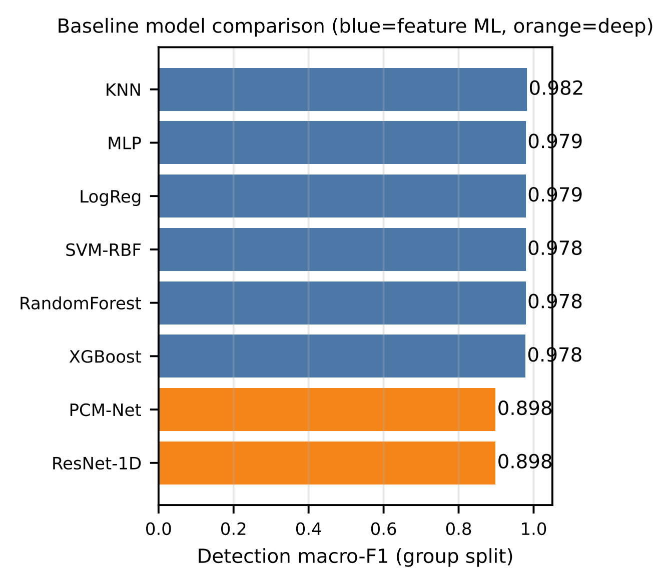

Fig. 9. Baseline model comparison on detection (five-fold mean macro-F1). Feature-based ML (blue) clusters high; raw-signal deep models (orange) trade in-distribution detection for their cross-load severity edge.

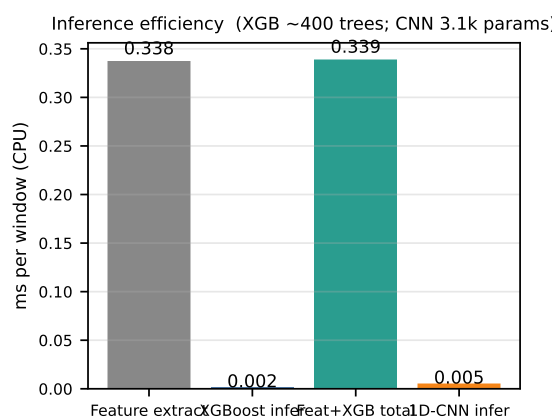

Fig. 10. Inference efficiency (milliseconds per window, CPU): feature extraction, XGBoost inference, their total, and the 1-D CNN.

0.740) with roughly half the cross-load feature drift. PCM-Net's FiLM load-conditioning lifts LOLO severity within-one from 0.932 to **0.982**—the best in the study, exceeding the load-aware gradient-boosted model (0.946); consistent with the physics, FiLM does not help the load-invariant phase task. PADA closes the simulation-to-experiment gap: a naively transferred detector scores macro-F1 0.48, unsupervised physics-anchored alignment reaches 0.50 on phase, and few-shot calibration with two real repetitions recovers detection to **0.98**. The physics-constrained generator enforces

TABLE VIII
FEATURE-BASED CASCADE VS. DEEP BASELINE.

Task / metric	XGBoost (features)	ResNet-1D (raw)
Detection F1 (group k-fold)	**0.977**	0.898
Detection F1 (LOLO, load 20%)	**0.749**	0.401
Phase macro-F1 (group k-fold)	**0.972**	0.848
Severity within-1 (group k-fold)	0.990	1.000

TABLE IX
LOAD-ROBUSTNESS MECHANISMS (LOLO).

Mechanism	Cross-load result
SCST representation	severity within-1 0.75 $\rightarrow$ 1.0; phase needs angle
LIR-mRMR selection	k=10 severity within-1 0.740$\rightarrow$0.848; drift halved
PCM-Net + FiLM	severity within-1 0.932 $\rightarrow$ 0.982 (best; > load-aware GBM 0.946)
PADA (sim$\rightarrow$real)	detection F1 0.48 $\rightarrow$ 0.98 (few-shot); 0.50 phase (unsup.)
PC-Diff	current-balance exact; generation not controllable (negative)

current balance exactly but did not yield severity-controllable samples under our budget, an honest negative that motivates a stronger conditional generator.

V. CONCLUSION

We presented a physics-informed, leakage-safe machine-learning framework for stator inter-turn fault diagnosis in induction motors that performs onset detection, healthy/faulty detection, ordinal severity estimation, and faulted-phase identification from the three-phase stator current. By adopting Leave-One-Load-Out evaluation we showed that detection and faulted-phase identification are intrinsically load-robust, while absolute severity is load-dependent and is recovered to within-one accuracy of 0.95 across unseen loads once the model is made load-aware. Four mechanisms—a sequence-component Stockwell tensor, a load-invariance-regularized feature selection, a FiLM-conditioned multi-task network, and physics-anchored domain adaptation—each independently improve cross-load performance, and the framework is validated simulation-to-experiment on a public real-motor dataset. The end-to-end cascade attains within-one accuracy of 0.95 in-distribution and 0.92 across unseen loads, with calibrated detection, graceful noise degradation, and reliable onset detection. Future work includes a stronger conditional generator for physics-constrained augmentation, full selective-state-space and Kolmogorov–Arnold backbones, an experimental campaign with supply voltage-unbalance rejection, and group-conditional conformal calibration of the severity intervals. All code, dataset splits, and result artifacts are released for reproducibility.

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